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XNC: XOR and NOT-Based Lossless Compression for Optimizing Unquantized Embedding Layers in Large Language Models

Junghyeok Lee, Jihoon Jang, and Hyun Kim

Department of Electrical and Information Engineering and Research Center for Electrical and Information Technology

Seoul National University of Science and Technology, Seoul, Korea

IDSL
Intelligent Digital Systems Design Lab.

SEOULTECH

Outline

- Motivation
- Background
- Contribution
- Proposed Method
- Experimental Results
- Conclusion

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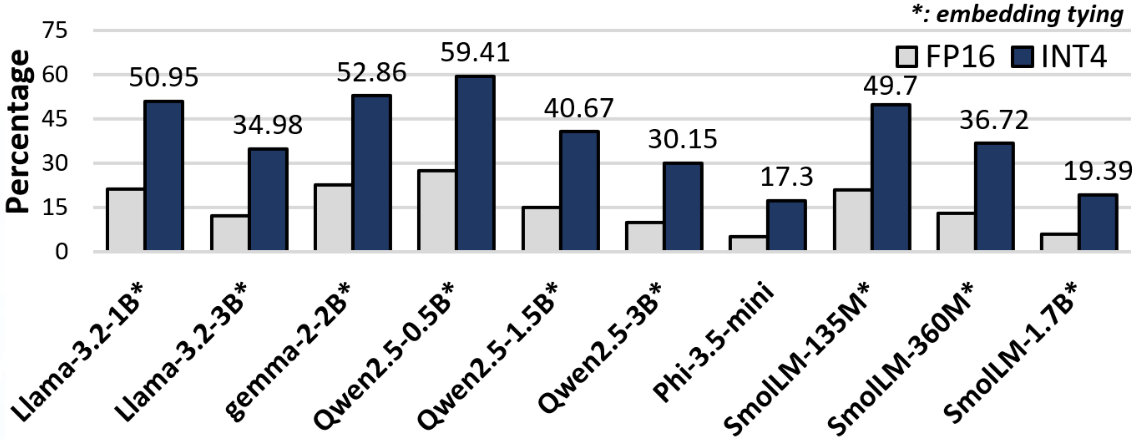
Motivation (1)

- **Challenges of hardware resource limitations in On-Device LLM deployment**
  - Compact LLM size (<4B parameters)
  - Model compression (quantization, pruning , and knowledge distillation)
- **Challenges in small LLM Compression**
  - Focus has been primarily on compressing attention and MLP layers
  - Increasing proportion of unquantized embedding layer parameters
  - Despite the release of small LLMs with embedding tying, model size remains large



<Parameter proportion in small LLMs>

Motivation (2)

- **Why the Embedding Layer Cannot Be Ignored?**
  - Reduction of decoder layers
  - Increase in vocabulary size
- **How can the embedding layer in small LLMs be further compressed?**
  - Lossy (quantization, pruning)
    → significant accuracy drop
  - Lossless (entropy encoding, run-length encoding)
    → difficult hardware implementation
    → high decompression cycle overhead

Need to simple hardware implementation & low decompression overhead!

Llama-2-7B	→ 3.0×	Llama-3.2-3B
[32000,4096]		[128256,3072]
Phi-3-mini	→ 6.2×	Phi-4-mini
[32064,3072]		[200064,3072]

<Embedding Layer Shape Trends in Small LLMs>

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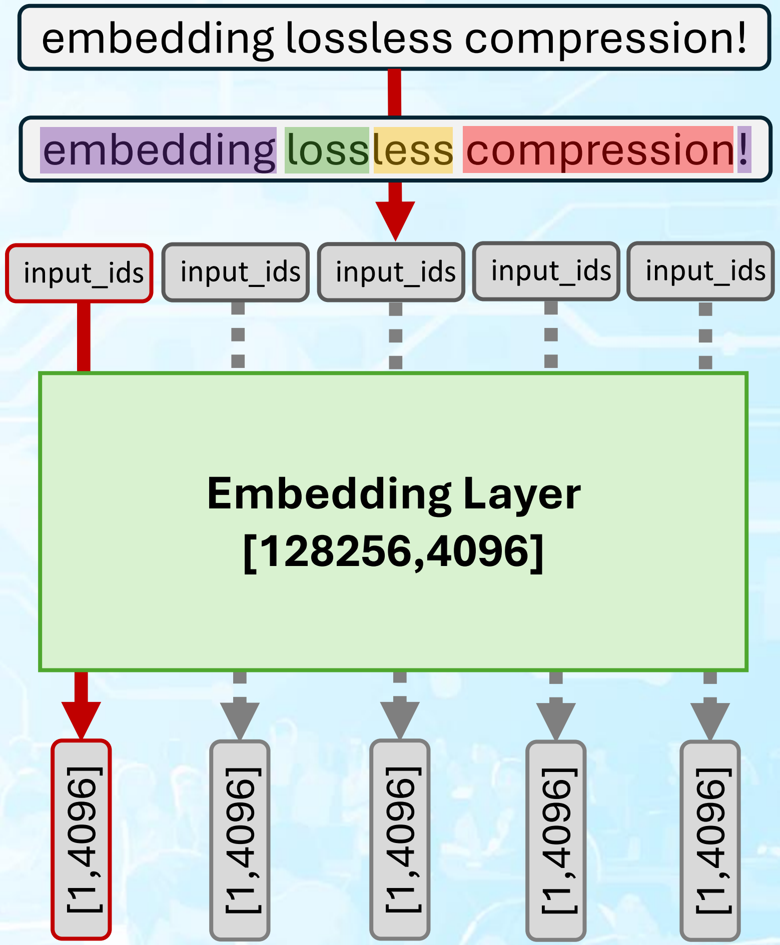
Background (1)

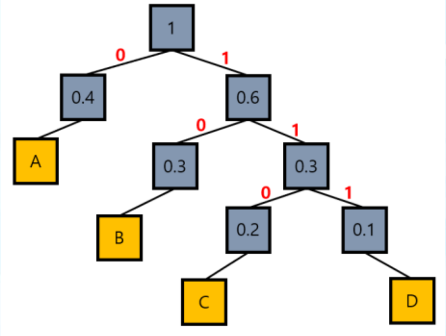
- **Embedding layer**
  - Embedding vectors capture semantic relationships between words
  - 16-bit precision is maintained, as small changes can degrade contextual understanding and prediction quality
- **Embedding layer operation**
  - Embed_token
    - lookup table operation
      - input_ids → embedding weight → input vector
  - Lm_head
    - GEMV operation
      - output vector · embedding weight = logits



<Example of embedding layer operation>

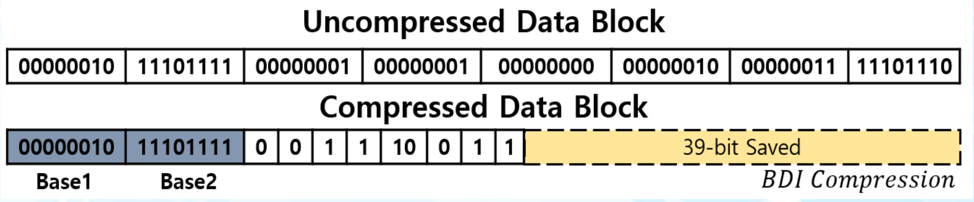
Background (2)

- **Lossless compression**
  - Compression based on statistical techniques such as Huffman coding and clustering
    - E^2MC [Lal+, IPDPS'17], SC^2 [Arelakis+, ISCA'14]
    - GBDI [Angerd+, HPCA'22]
  - Compression leveraging frequent patterns in data
    - FPC [Alamedeen+, UW-Madison TR'04]
    - C-PACK [Chen+, TVLSI'09]
  - Base-delta compression leveraging differences between data values
    - BDI [Pekhimenko+, PACT'12], BPC [Kim+, ISCA'16]
    - GBDI [Angerd+, HPCA'22]
  - Run-length encoding for efficiently compressing values made consecutive through transformation
    - BPC [Kim+, ISCA'16]

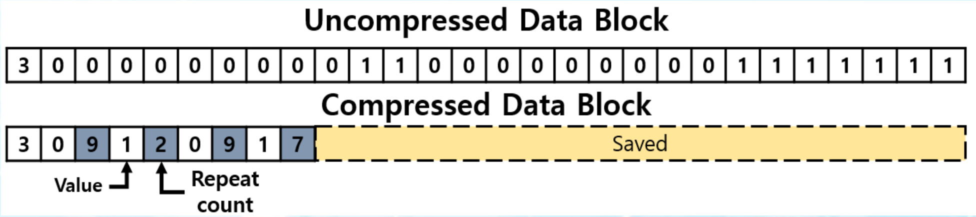


Symbol	Probability	encoding
A	0.4	0
B	0.3	10
C	0.2	110
D	0.1	111

<Huffman coding>



<Base-delta compression>



<Run-length encoding>

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Contribution

- **Propose a lossless compression method for efficient LLM embedding layers**
  - Reduce metadata overhead using block-wise metadata
- **Improve compression ratio by increasing frequent pattern occurrence through embedding layer transformation**
- **Hardware-friendly compressor and decompressor using XOR and NOT operations**
  - Simple hardware logic
  - Low compressor, decompressor cycle

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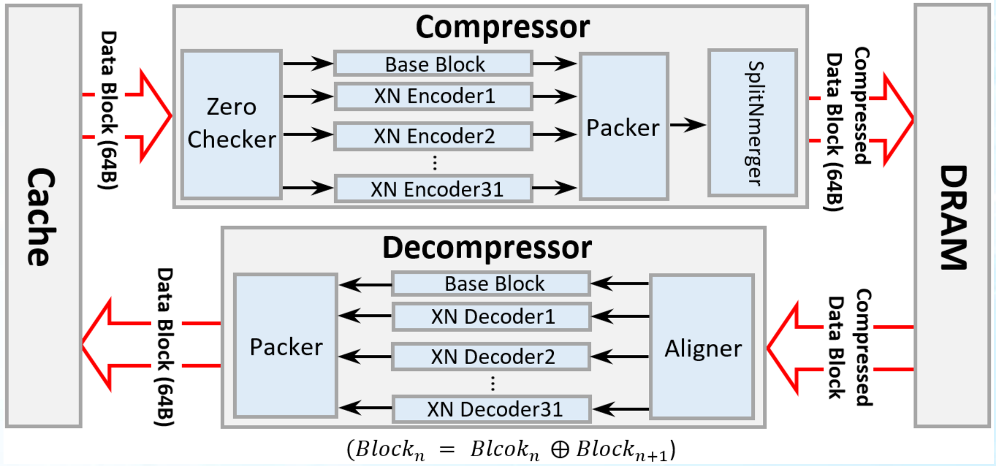
Proposed Method (1)

- **Compressor**
  - Zero checker
    - → data block type metadata 1-bit
  - Encoder
    - → encoding type metadata 1-bit
  - Packer
    - → 33-bit metadata + 32 compressed values
  - SplitNmerger
    - → reconstruct into a 64B data block
- **Decompressor**
  - Aligner
    - → check the 1-bit metadata for the data block type
  - Decoder
    - → check the 32-bit encoding type for the 32 compressed values
  - Packer
    - → 32 decompressed values



<Overview of proposed XNC>

Proposed Method (2)

- **Compression method**
  1. Zero checker
     - check if specific bits (e.g., 2nd, 14th, 15th, and 16th)
       → per 64B data block

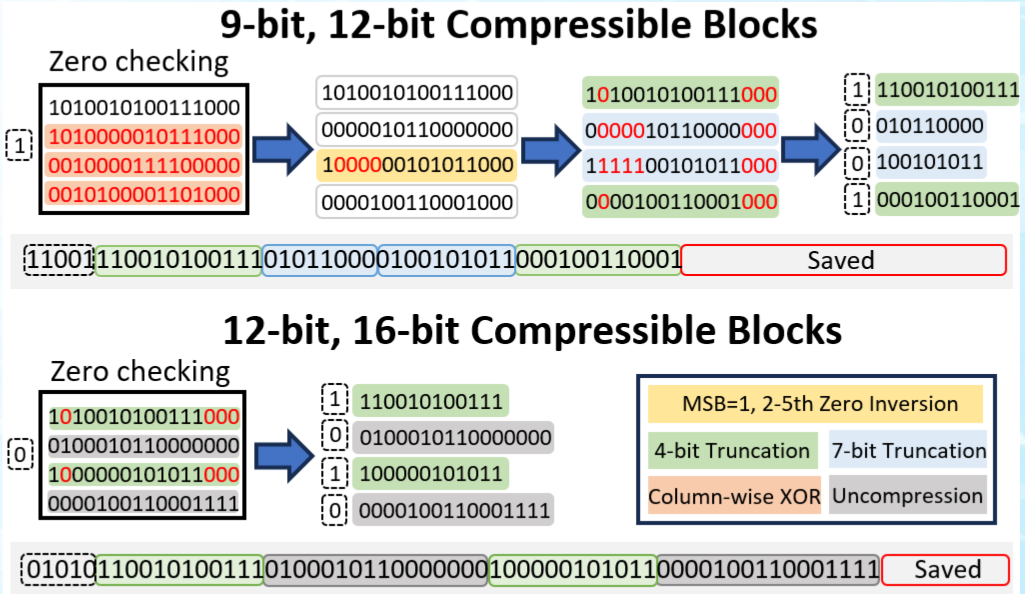
  2-1. 9-bit, 12-bit compressible data block
     - if MSB = 1, invert bits 2 to 5
       → no metadata needed (MSB preserved)
     - apply 4-bit or 7-bit truncation

  2-2. 12-bit, 16-bit compressible data block
     - apply 4-bit truncation
     - preserve 16-bit format

  3. Packer
     - a 33-bit metadata is stored as the header of the compressed data block.

	Llama-3.2-1B	Gemma-2-2B	Qwen2.5-0.5B	Phi-3.5-mini	SmoLM-135M
All zero	≈ 0%	≈ 0%	≈ 0%	≈ 0%	0%
Narrow	0.26%	0.26%	0.26%	0.27%	0.26%
Non-pattern	99.73%	99.73%	99.73%	99.72%	99.74%
Specific bits all zero	99.83%	99.90%	99.79%	99.75%	99.96%
Specific bits Non-all zero	0.17%	0.10%	0.21%	0.25%	0.04%

<Data pattern of FP16 embedding layer weights>



<Example of XNC Transformation & Compression>

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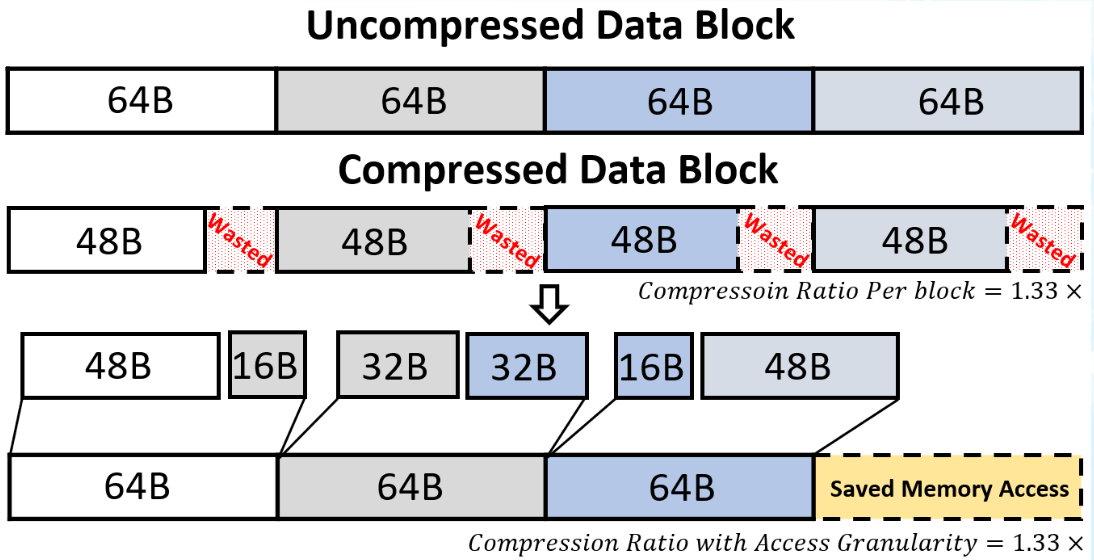
Proposed Method (3)

- **How to leverage wasted space after compression?**
  - If unused space isn't utilized during data transfer, compression loses its benefit
  - The SplitNmerger module of XNC reconstructs 48B compressed blocks into 64B blocks to minimize waste
  - Reduces memory access and saves bandwidth for efficient transfer



<Example of memory access savings through compression>

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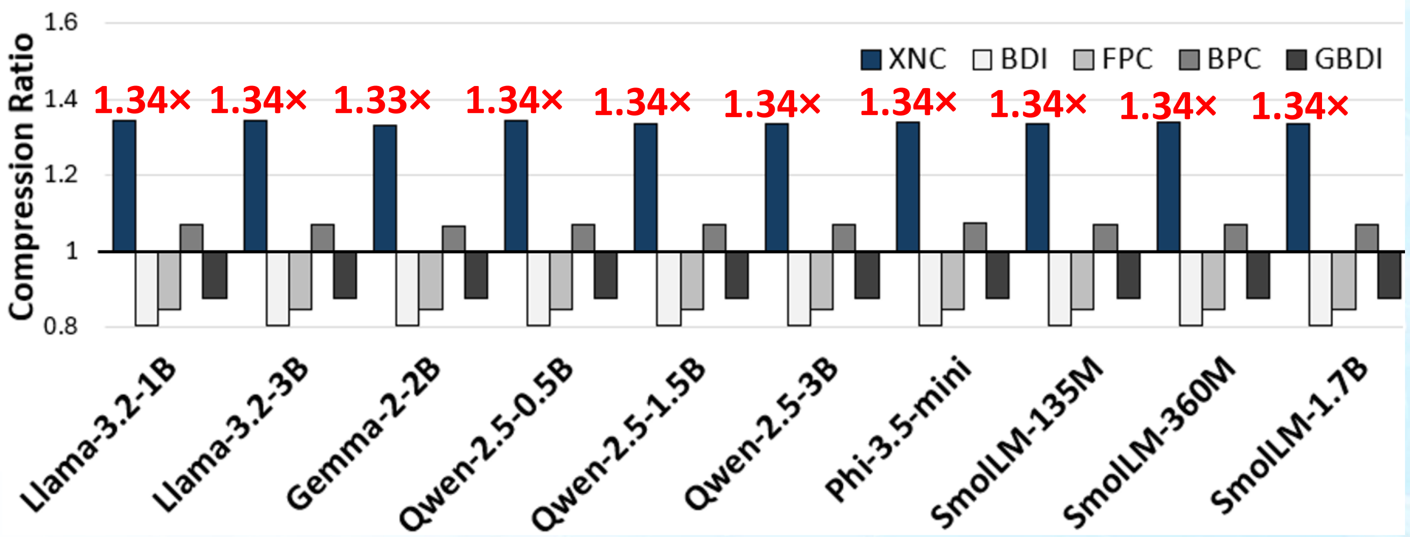
Experimental Results (1)

- **Experimental environments**
  - Pre-trained Hugging Face models under 4B parameters
  - Compared compression methods are adjusted to 16-bit precision
  - Metadata is included in compression ratio evaluation
- **Compression ratio**
  - The highest average compression ratio at 1.34x



<Compression Ratio Comparison of Embedding Layers in small LLMs>

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Experimental Results (2)

- **Compression/decompression cycle**
  - Fewer compression/decompression cycles than other lossless methods (except BDI)
- **Compression of Small LLMs and VLMs**
  - Average of 9.68% lossless compression on 4-bit quantized small LLMs

Model	4-bit Quantized Model Size (MB)	Model Size after XNC (MB)	Compression Percent (%)
Llama-3.2-1B	1031	897	13.0
Llama-3.2-3B	2253	2052	8.9
Gemma-2-2B	2232	1939	13.2
Qwen2.5-0.5B	458	388	15.3
Qwen2.5-1.5B	1147	1030	10.2
Qwen2.5-3B	2064	1907	7.6
Phi-3.5-mini	2277	2177	4.4
SmolLM-135M	114	100	12.3
SmolLM-350M	257	233	9.3
SmolLM-1.7B	1038	987	4.9
SmolVLM-256M	299	271	9.4
Qwen2.5-VL-3B	3226	3069	4.9

<Compression Percentage in LLMs with Applied XNC Method>

Technique	BDI	FPC	BPC	GBDI	XNC
Compressor Latency (Cycle)	1	9	9	12	4
Decompressor Latency (Cycle)	1	9	9	9	3

<Comparison of Cycle for Compressor and Decompressor>

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Conclusion

- **Propose a lossless compression method for small LLM embedding layers**
  - Effectively truncates frequently occurring upper and lower bits after data transformation
- **Hardware-friendly compression and decompression engine**
  - 3-cycle decompression latency
- **Compresses embedding layers by 1.34× on average, reducing memory accesses**

Q & A

Thank you!

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